



Mixed-precision algorithms for Sylvester equations

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Sylvester matrix equation

Given $A \in \mathbb{C}^{m \times m}$, $B \in \mathbb{C}^{n \times n}$, and $C \in \mathbb{C}^{m \times n}$, compute the solution $X \in \mathbb{C}^{m \times n}$ to

$$AX + XB = C.$$

The equation can be recast as the $mn \times mn$ linear system

$$M \text{vec}(X) = c, \quad M := I_n \otimes A + B^T \otimes I_m, \quad c := \text{vec}(C).$$

Applications include, theoretically, **block diagonalization** and **perturbation theory for matrix functions**, and, practically, many in, e.g., **system balancing**, **control**, and **model reduction**.

Bartels–Stewart algorithm

- 1: Compute the **Schur decompositions** $A =: U_A \hat{T}_A U_A^*$ and $B =: U_B \hat{T}_B U_B^*$.
- 2: Multiplying the Sylvester equation by U_A^* on the left and by U_B on the right to obtain the equation

$$\hat{T}_A Y + Y \hat{T}_B = \tilde{C}, \quad \tilde{C} := U_A^* C U_B.$$

- 3: Solve the (upper) quasi-triangular Sylvester equation by the recurrence (working up the block columns from first to last)

$$\hat{T}_A(i, i) Y_{ij} + Y_{ij} \hat{T}_B(j, j) = \tilde{C}_{ij} - \sum_{k=i+1}^m \hat{T}_A(i, k) Y_{kj} - \sum_{k=1}^{j-1} Y_{ik} \hat{T}_B(k, j)$$

with Y partitioned conformally with $\hat{T}_A$ and $\hat{T}_B$.

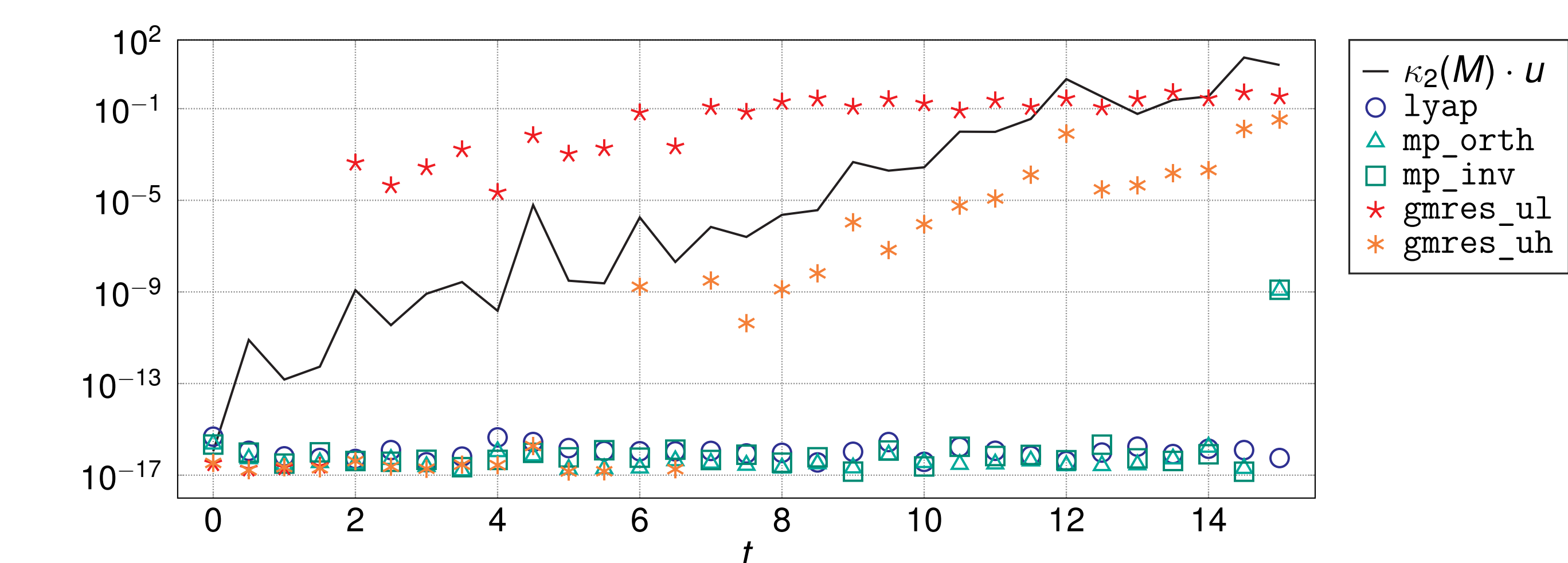
- 4: Recover solution to the full Sylvester equation $X := U_A Y U_B^*$.

Schur-left-preconditioned GMRES-IR for Sylvester equation

- Key step: apply preconditioner $M^{-1} = (\hat{U}_B \otimes U_A)(I_n \otimes \hat{T}_A + \hat{T}_B^T \otimes I_m)^{-1}(\hat{U}_B^T \otimes U_A^*)$ implicitly to vectors.
- In our *two-precision* setting, $u_g = u_\ell$ (gmres_ul) or $u_g = u_h$ (gmres_uh).

- 1: Compute Schur decompositions $A =: \hat{U}_A \hat{T}_A \hat{U}_A^*$ and $B =: \hat{U}_B \hat{T}_B \hat{U}_B^*$ in precision u_ℓ .
- 2: Initialize X_0 (to, e.g., $X_0 \leftarrow 0$), $i \leftarrow 0$
- 3: **repeat**
- 4: Compute $R_i = C - AX_i - X_i B$ in precision u_h .
- 5: $R_i \leftarrow \hat{U}_A^* R_i \hat{U}_B$ in precision u_g .
- 6: $R_i \leftarrow$ solution X of $\hat{T}_A X + X \hat{T}_B^T = R_i$ in precision u_g .
- 7: $R_i \leftarrow \hat{U}_A R_i \hat{U}_B^*$ in precision u_g . $\triangleright M^{-1}$ applied to the residual at precision u_g
- 8: Solve $M^{-1} M \text{vec}(E_i) = \text{vec}(R_i)$ by GMRES in precision u_g with $M^{-1}(M(\cdot))$ applied to the vector implicitly in precision u_h , where $M = I_n \otimes A + B^T \otimes I_m$.
- 9: $X_{i+1} \leftarrow X_i + E_i$ in precision u_h , $i \leftarrow i + 1$
- 10: **until** converged

Comparison of MATLAB `lyap` and our MATLAB implementations of the mixed-precision algorithms on pseudorandom problems with $\kappa_2(M) \approx 10^4$. We use $u_\ell = 2^{-24}$ (binary32) and $u_h = 2^{-53}$ (binary64) throughout.



- Accuracy of gmres_ul and gmres_uh deteriorates as condition number of the equation increases.
- For gmres_ul and gmres_uh to be competitive in terms of **efficiency**, the GMRES solver on Line 8 at each outer iteration has to converge quickly.

Iterative refinement for quasi-triangular equation

- Use: obtain a working-precision solution to the equation $(\hat{T}_A + \Delta \hat{T}_A)Y + Y(\hat{T}_B + \Delta \hat{T}_B) = F$ by using the low-precision Schur factors $\hat{T}_A$ and $\hat{T}_B$ in a stationary iteration ($\Delta \hat{T}_A$ and $\Delta \hat{T}_B$ are unstructured).

Input: $\hat{T}_A$, $\Delta \hat{T}_A$, $\hat{T}_B$, $\Delta \hat{T}_B$, and F . Initial guess Y_0 and target tolerance $\varepsilon > 0$.

Output: $Y \in \mathbb{C}^{m \times n}$ such that $(\hat{T}_A + \Delta \hat{T}_A)Y + Y(\hat{T}_B + \Delta \hat{T}_B) \approx F$.

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1 function SOLVE_PERT_SYLV_TRI_STAT( $\hat{T}_A$ ,  $\Delta \hat{T}_A$ ,  $\hat{T}_B$ ,  $\Delta \hat{T}_B$ ,  $F$ ,  $Y_0$ )
2    $i \leftarrow 0$ 
3   repeat
4     Find  $E_i$  such that  $\hat{T}_A E_i + E_i \hat{T}_B = F - (\hat{T}_A + \Delta \hat{T}_A)Y_i - Y_i(\hat{T}_B + \Delta \hat{T}_B)$ .
5      $Y_{i+1} \leftarrow Y_i + E_i$ 
6      $i \leftarrow i + 1$ 
7   until  $\|E_{i-1}\|/\|Y_i\| \leq \varepsilon$ 
8   return  $Y_i$ 
```

- SOLVE_PERT_SYLV_TRI_STAT is a **building block** of the two mixed-precision algorithms.
- Sufficient condition for **convergence**: $\|\Delta \hat{T}_A\|_2 + \|\Delta \hat{T}_B\|_2 < \text{sep}_F(\hat{T}_A, -\hat{T}_B)$.
- **Rounding error analysis** done by connecting SOLVE_PERT_SYLV_TRI_STAT to a fixed-precision iterative refinement scheme for linear systems.

Reference

R. H. Bartels and G. W. Stewart, *Algorithm 432: Solution of the matrix equation $AX + XB = C$* , Comm. ACM, 15 (1972), pp. 820–826.

Mixed-precision algorithms for $AX + XB = C$

- **Key point: obtain unitary Schur factors that are orthonormal to the working precision, in order to recover the solution of the original equation from the quasi-triangular equation.**

i. Algorithm `mp_orth`: By reorthonormalizing the low-precision unitary factors in working precision.

- 1: Compute Schur decompositions $A =: \hat{U}_A \hat{T}_A \hat{U}_A^*$, $B =: \hat{U}_B \hat{T}_B \hat{U}_B^*$ in precision u_ℓ .
- 2: Compute QR decompositions $U_A =: Q_A R_A$, $U_B =: Q_B R_B$, with positive diagonal, in precision u_h .
- 3: $\Delta \hat{T}_A \leftarrow Q_A^* A Q_A - \hat{T}_A$, computed in precision u_h .
- 4: $\Delta \hat{T}_B \leftarrow Q_B^* B Q_B - \hat{T}_B$, computed in precision u_h .
- 5: $F \leftarrow Q_A^* C Q_B$, computed in precision u_h .
- 6: Find Y_0 such that $\hat{T}_A Y_0 + Y_0 \hat{T}_B = F$ in precision u_ℓ .
- 7: $Y \leftarrow \text{SOLVE_PERT_SYLV_TRI_STAT}(\hat{T}_A, \Delta \hat{T}_A, \hat{T}_B, \Delta \hat{T}_B, F, Y_0)$
- 8: $X \leftarrow Q_A Y Q_B^*$

- Can prove the equation that `mp_orth` is solving differs from the original by $O(u_h)$ perturbations.
- Sufficient **convergence** condition: $\|\Delta \hat{T}_A\| \lesssim u_\ell(2\|\hat{T}_A\| + \kappa(Q_A)\|A\|)$, $\|\Delta \hat{T}_B\| \lesssim u_\ell(2\|\hat{T}_B\| + \kappa(Q_B)\|B\|)$.

ii. Algorithm `mp_inv`: By inverting the low-precision unitary factors in the working precision.

- 1: Compute Schur decompositions $A =: \hat{U}_A \hat{T}_A \hat{U}_A^*$, $B =: \hat{U}_B \hat{T}_B \hat{U}_B^*$ in precision u_ℓ .
- 2: Compute LU decompositions of $\hat{U}_A^*$ and $\hat{U}_B$ in precision u_h .
- 3: $\Delta \hat{T}_A \leftarrow \hat{U}_A^* A \hat{U}_A^{-*} - \hat{T}_A$, computed in precision u_h .
- 4: $\Delta \hat{T}_B \leftarrow \hat{U}_B^{-1} B \hat{U}_B - \hat{T}_B$, computed in precision u_h .
- 5: $F \leftarrow \hat{U}_A^* C \hat{U}_B$, computed in precision u_h .
- 6: Find Y_0 such that $\hat{T}_A Y_0 + Y_0 \hat{T}_B = F$ in precision u_ℓ .
- 7: $Y \leftarrow \text{SOLVE_PERT_SYLV_TRI_STAT}(\hat{T}_A, \Delta \hat{T}_A, \hat{T}_B, \Delta \hat{T}_B, F, Y_0)$
- 8: $X \leftarrow \hat{U}_A^{-*} Y \hat{U}_B^{-1}$

- Note that the right-hand side F and the solution Y of the quasi-triangular equations are different in `mp_orth` and `mp_inv`.
- Sufficient **convergence** condition: $\|\Delta \hat{T}_A\| \lesssim u_\ell(\|\hat{T}_A\| + \kappa(\hat{U}_A^*)\|A\|)$, $\|\Delta \hat{T}_B\| \lesssim u_\ell(\|\hat{T}_B\| + \kappa(\hat{U}_B)\|B\|)$.

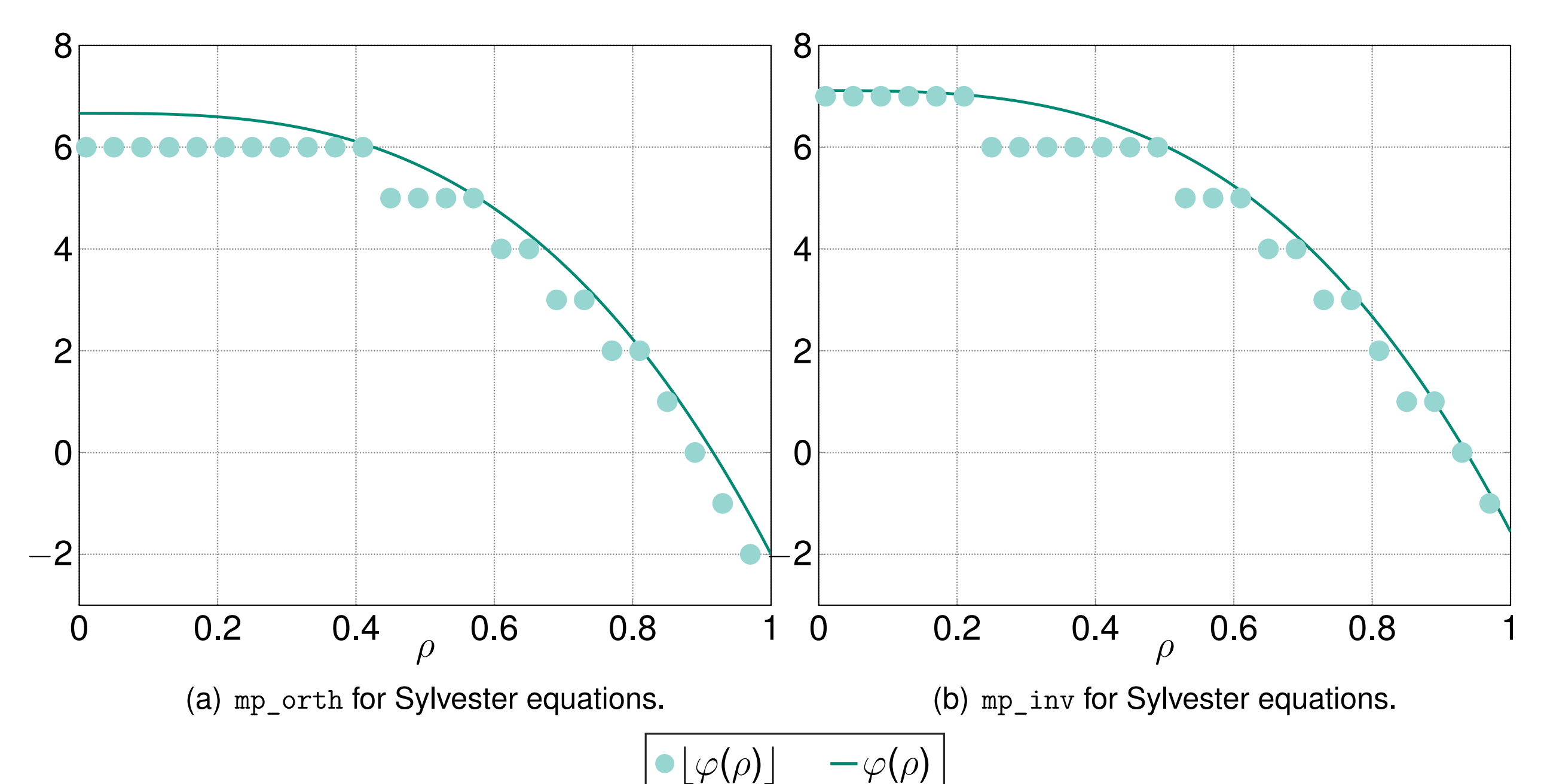
Computational cost and comparison model

The three constants are $\alpha := m^3 + n^3$, $\beta := mn(m + n)$, and $\gamma := n^3$.

	Sylvester		Lyapunov ($B = A^*$)	
	u_ℓ flops	u_h flops	u_ℓ flops	u_h flops
lyap	-	$25\alpha + 5\beta$	-	35γ
mp_orth	$25\alpha + \beta$	$6\alpha + (4 + 3k)\beta$	27γ	$(14 + 6k)\gamma$
mp_inv	$25\alpha + \beta$	$4\frac{2}{3}\alpha + (4 + 3k)\beta$	27γ	$(12\frac{2}{3} + 6k)\gamma$

- **When will the mixed-precision algorithms be faster than the Bartels–Stewart algorithm in u_h ?**

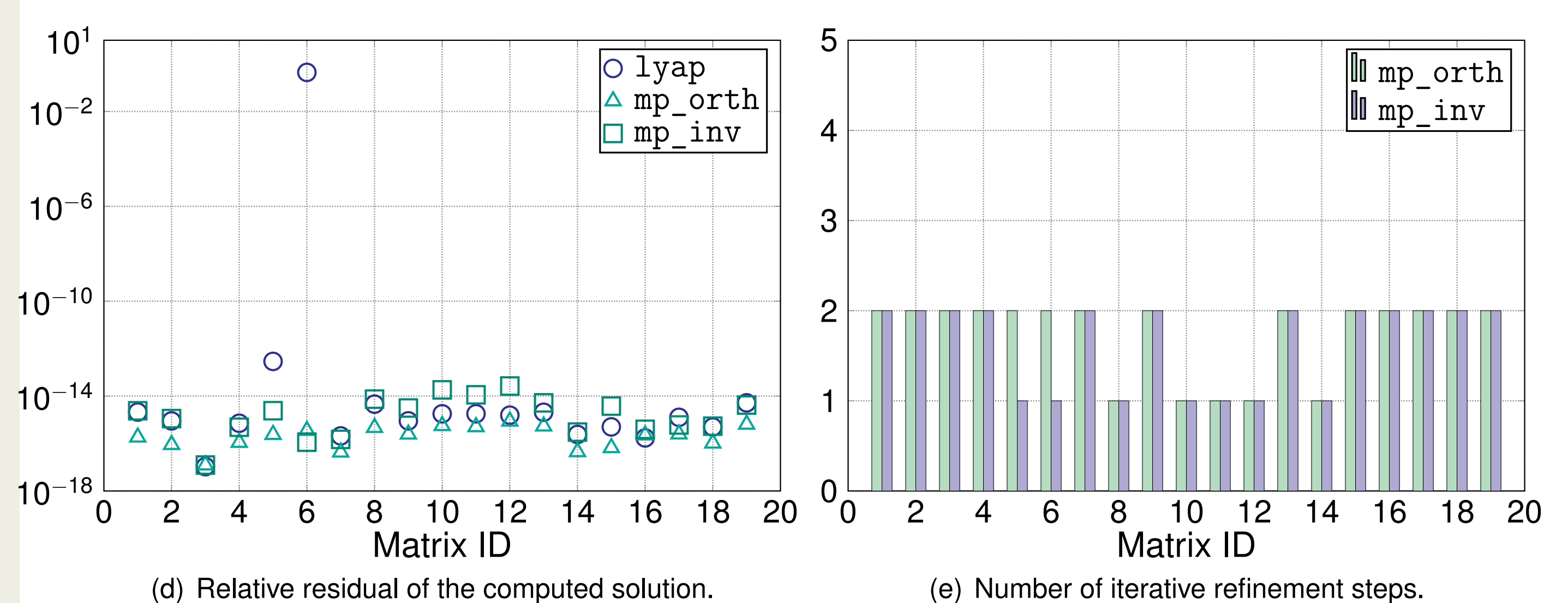
Maximum number of iterations against the ratio ρ of the computational cost of a flop in precision u_ℓ to one in precision u_h , for which the mixed-precision algorithms will be asymptotically faster than the Bartels–Stewart algorithm run in high precision.



- $\varphi(\rho)$ is plotted with $m = n$; $\rho \leq 0.5$ in practical interest (e.g., $\rho \approx 0.5$ for binary32/binary64).
- For **Lyapunov** the trend is similar, though relatively fewer iterations are allowed.

Experiments on real-world datasets

Convergence tolerance is $10^{-15} \max(m, n)$ with Frobenius norm. Test matrices have sizes between 31×31 and 1668×1668 .



- For Lyapunov the trend is similar and typically **two steps** are performed for convergence.
- The new algorithms are faster than Bartels–Stewart if $\rho \leq 0.5$ for binary32/binary64.